

Chapter 4

Image and Video Analysis

IMP302m

Contents

4.1 Image transforms

4.2 Discrete Hartley transform

4.3 Discrete Cosine transform

4.4 Walsh-Hadamard Transforms

4.5 Wavelet transform

4.6 Multiscale Image Decompositions and Wavelets

4.7 Machine learning based Image and Video analysis

4.1 Image transforms

- Image transforms:
 - Decompose functions into weighted sums of orthogonal or biorthogonal basis functions.
 - Studied using linear algebra and functional analysis.
- Images are treated as vectors in the vector space of all images: Basis functions define the nature and utility of image transforms
- Properties
 - Transforms represent the coefficients of linear expansions.
 - All transforms of an image contain the same information and total energy.
 - Transforms are reversible and differ only in the distribution of information and energy among coefficients.

4.2 Discrete Hartley transform

- 2D Discrete Hartley transform of a 2D signal or image $f(x,y)$ of size $N \times M$ is given by

$$F_{\text{DHT}}(u, v) = \sum_{x=0}^{N-1} \sum_{y=0}^{M-1} f(x, y) \text{cas} \left(\frac{2\pi}{N} ux + \frac{2\pi}{M} vy \right)$$

where $\text{cas}(x) = \cos(x) + \sin(x)$

4.2 Discrete Hartley transform – Properties

- Real-valued output: produces real-valued results, simplifying computations
- Symmetry: The 2D DHT is symmetric, meaning $H(u, v) = H(-u, -v)$
- Energy conservation: The DHT preserves the total energy of the signal, similar to the DFT.
- The inverse of the 2D DHT can be used to reconstruct the original image or signal from its transform, making it a reversible process
- DHT transformation matrix: real, orthogonal, and symmetric

4.2 Discrete Hartley transform – Implementation

- $N = 4$

4.3 Discrete Cosine transform

- 2D Discrete Cosine transform of a 2D signal or image $f(x,y)$ of size $N \times M$ is given by

$$F_{\text{DCT}}(u, v) = \alpha(u)\alpha(v) \sum_{x=0}^{N-1} \sum_{y=0}^{M-1} f(x, y) \cos \left[\frac{(2x+1)u\pi}{2N} \right] \cos \left[\frac{(2y+1)v\pi}{2M} \right]$$

- where $\alpha(u)$ and $\alpha(v)$ are normalization factors given by

$$\alpha(k) = \begin{cases} \sqrt{\frac{1}{N}} & \text{if } k = 0 \\ \sqrt{\frac{2}{N}} & \text{if } k \neq 0 \end{cases}$$

4.3 Discrete Cosine transform

- The spectrum of a discrete cosine transform has roughly the same frequency range as that of the Fourier and Hartley transforms, but twice the frequency resolution
- Rather than N -point periodicity, the underlying assumption of the DFT, the discrete cosine transform assumes $2N$ -point periodicity and even symmetry
- $2N$ -point periodicity and even symmetry minimize discontinuity, as well as the accompanying high-frequency artifact → important advantage of the DCT in image compression

4.3 Discrete Cosine transform— Implementation

- $N = 4$

4.3 Discrete Cosine transform – Discrete Sine transform

- 2D Discrete Sine transform of a 2D signal or image $f(x,y)$ of size $N \times M$ is given by

$$F_{\text{DST}}(u, v) = \frac{2}{\sqrt{(N+1)(M+1)}} \sum_{x=0}^{N-1} \sum_{y=0}^{M-1} f(x, y) \sin \left[\frac{(x+1)(u+1)\pi}{N+1} \right] \cos \left[\frac{(y+1)(v+1)\pi}{M+1} \right]$$

4.3 Discrete Cosine transform – Discrete Sine transform

- Like the DCT, the DST has roughly the same frequency range as the DFT, but twice the frequency resolution
- Unlike both the DCT and DFT, the DST has no DC (at $u = 0$) component. This results from an underlying assumption that the function being transformed is $2(N+1)$ -point periodic and odd symmetric, making its average value zero
- DST does not reduce boundary discontinuity

4.4 Walsh-Hadamard Transform

- Non-sinusoidal transformations that decompose a function into a linear combination of rectangular basis functions, called Walsh functions, of value $+1$ and -1
- It decomposes a signal into a set of orthogonal square waveforms, offering an alternative to Fourier-based transforms

4.4 Walsh-Hadamard Transform

- 2D Walsh-Hadamard transform of a 2D signal or image $f(x,y)$ of size $N \times N$ is given by

$$F_{\text{WHT}}(u, v) = \mathbf{H}_N \cdot f(x, y) \cdot \mathbf{H}_N^T$$

where $\mathbf{H}_N$ is the Hadamard matrix of size N .

$$\mathbf{H}_1 = [1] \quad \mathbf{H}_2 = \begin{bmatrix} \mathbf{H}_1 & \mathbf{H}_1 \\ \mathbf{H}_1 & -\mathbf{H}_1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$\mathbf{H}_{2n} = \begin{bmatrix} \mathbf{H}_n & \mathbf{H}_n \\ \mathbf{H}_n & -\mathbf{H}_n \end{bmatrix}$$

4.4 Walsh-Hadamard Transform

- The inverse 2D WHT is identical to the forward transform, making it easy to reconstruct the original image:

$$f(x, y) = \frac{1}{N^2} \mathbf{H}_N \cdot F_{\text{WHT}}(u, v) \cdot \mathbf{H}_N^T$$

- The transform is lossless and fully reversible.

4.4 Walsh-Hadamard Transform – Properties

- Orthogonality: The 2D WHT basis functions are orthogonal, ensuring that the transformation preserves the signal's information and energy.
- Binary operations: The 2D WHT uses only addition and subtraction, making it efficient for digital implementations.
- Computational Efficiency: The Fast Walsh-Hadamard Transform (FWHT) can be applied in 2D, reducing the computational complexity to $O(NM \log(NM))$, similar to the 2D Fast Fourier Transform (FFT).

4.4 Walsh-Hadamard Transform – Applications

- Image Compression: The 2D WHT is used in image compression due to its ability to compact energy into a few coefficients, similar to the Discrete Cosine Transform (DCT) but with simpler computations.
- Pattern Recognition: Used for feature extraction and recognition tasks by analyzing the frequency content of 2D binary signals or images.
- Digital Signal Processing (DSP): Applied in the analysis and processing of 2D signals where fast, low-complexity transforms are required.

4.5 Wavelet transform

- The foundation of a powerful approach to signal processing and analysis called multiresolution theory
- Multiresolution theory incorporates and unifies techniques from a variety of disciplines, including subband coding from signal processing, quadrature mirror filtering from digital speech recognition, and pyramidal image processing
- It is concerned with the representation and analysis of signals (or images) at more than one resolution

4.5 Wavelet transform - Discrete Wavelet transform

- A set of basis functions composed of all the real father scaling function $\varphi(x)$

$$\varphi_{j,k}(t) = \sqrt{2^j} \varphi(2^j t - k)$$

where translation k and scale (dilation) j

- The scaling function is orthogonal to its integer translates

$$\varphi(t) = \sum_{k \in \mathbb{Z}} h_{\varphi}(k) \varphi_{j,k}(t) = \sum_{k \in \mathbb{Z}} h_{\varphi}(k) \sqrt{2} \varphi(2t - k)$$

Scaling function coefficients

- and

$$h_{\varphi}(k) = \langle \varphi(t), \sqrt{2} \varphi(2t - k) \rangle = \int_{-\infty}^{\infty} \varphi^*(t) \sqrt{2} \varphi(2t - k) dt$$

4.5 Wavelet transform – Discrete Wavelet transform

- $\varphi(x)$ meets the MRA requirements of the previous section, there exists a *mother wavelet function* $\psi(t)$ whose integer translations and binary scalings

$$\psi_{j,k}(t) = \sqrt{2^j} \psi(2^j t - k)$$

integer translation k and scale j

- Wavelet space resides within scaling space

$$\psi(t) = \sum_{k \in \mathbb{Z}} h_{\psi}(k) \sqrt{2} \varphi(2t - k)$$

← wavelet function coefficients

- Wavelet function and scaling function coefficients are related

$$h_{\psi}(k) = (-1)^k h_{\varphi}(1 - k)$$

4.5 Wavelet transform – Discrete Wavelet transform

- Discrete Wavelet transform decomposes the signal f into

$$f(t) = \sum_k c_{j_0}(k) \varphi_{j_0,k}(t) + \sum_{j=0}^{\infty} \sum_k d_j(k) \psi_{j,k}(t)$$

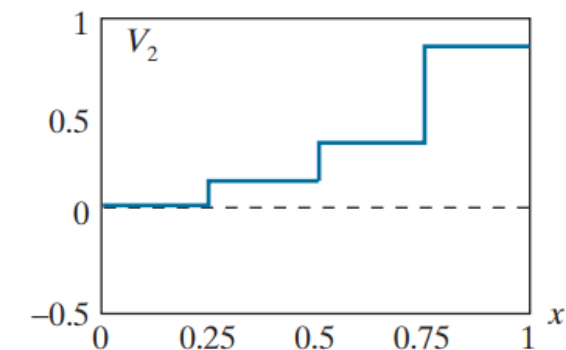
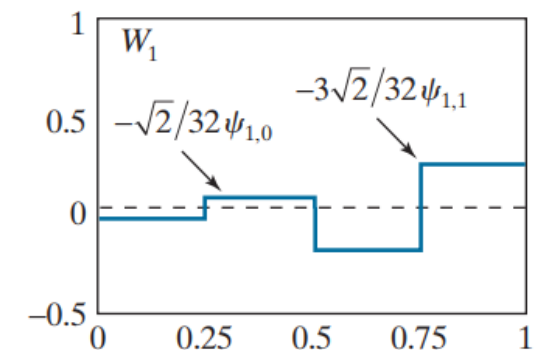
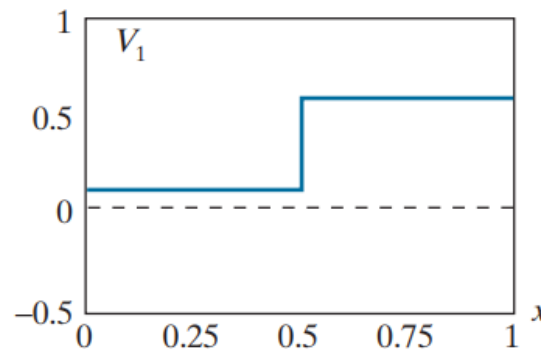
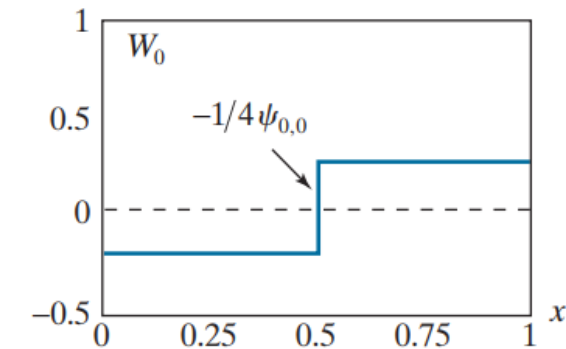
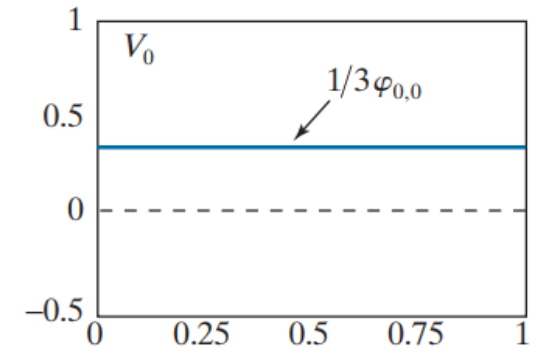
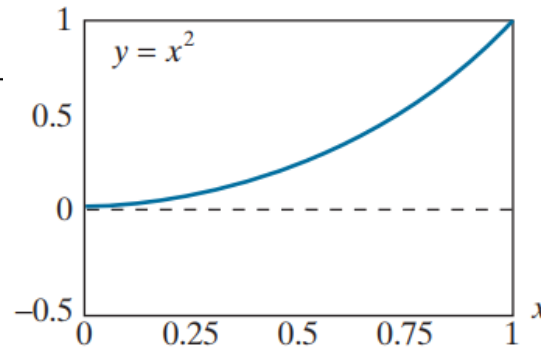
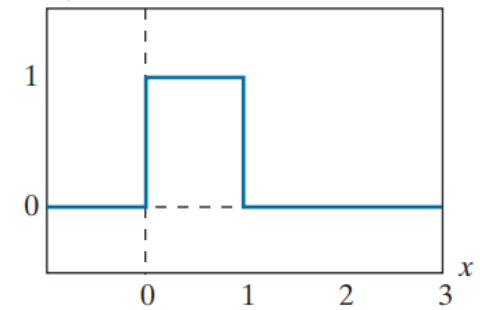
where c_{j_0} and d_j for $j \geq j_0$ are called *approximation* and *detail coefficients*. If the scaling and wavelet functions are orthonormal

$$c_{j_0} = \langle f(t), \varphi_{j_0,k}(t) \rangle \quad d_j = \langle f(t), \psi_{j,k}(t) \rangle$$

4.5 Wavelet transform – Continuous Wavelet transform

- The unit-height, unit-width scaling function $\varphi(x) = \begin{cases} 1 & \text{if } 0 \leq x < 1 \\ 0 & \text{otherwise} \end{cases}$
- Wavelet series expansion of the following function

$$y = \begin{cases} x^2 & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

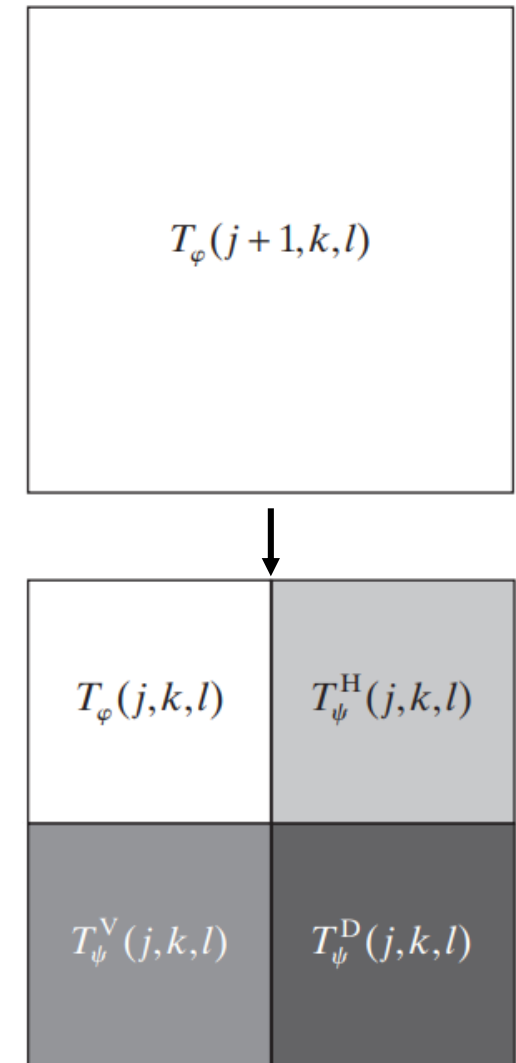
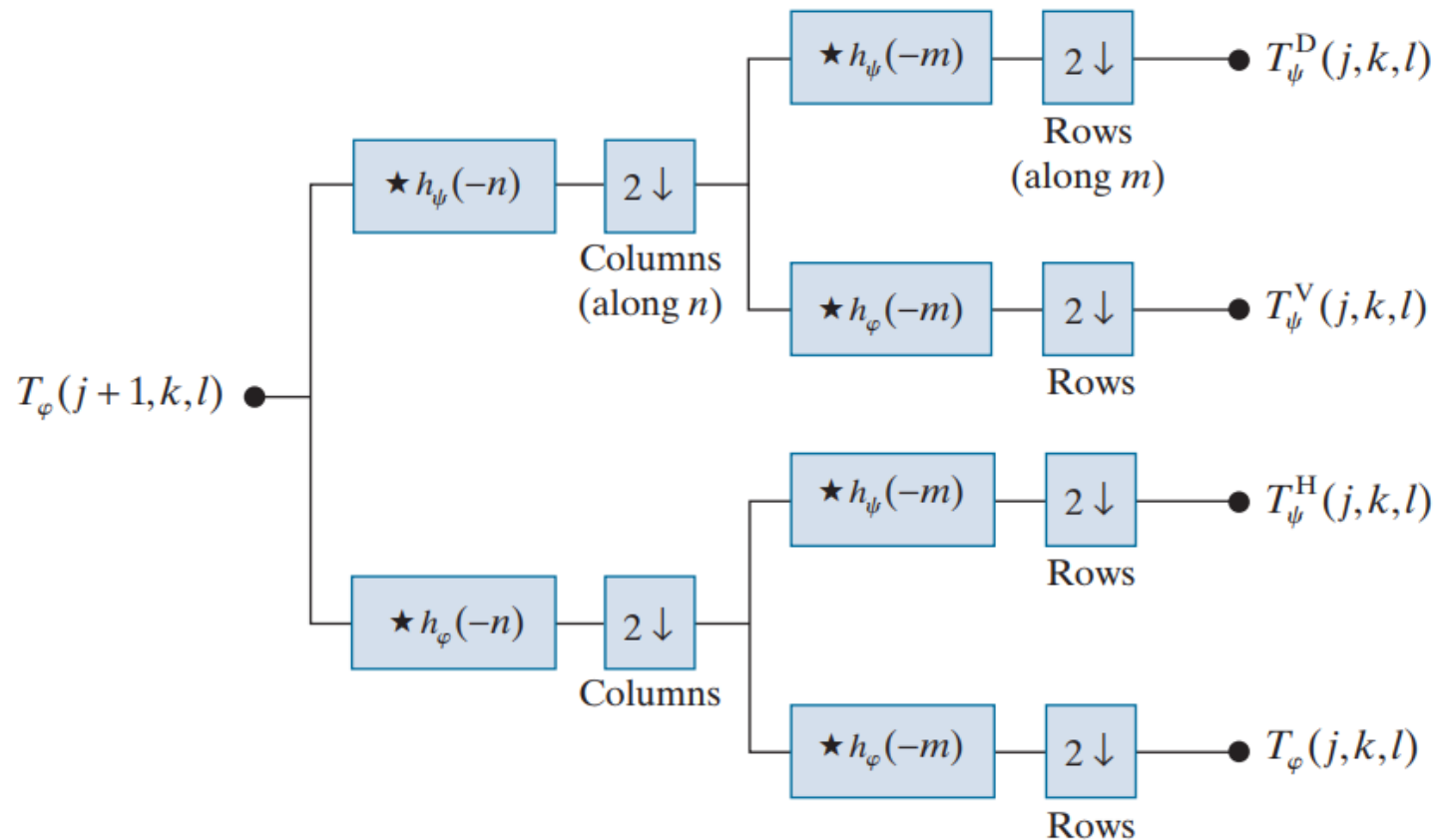


4.6 Multiscale Image Decompositions and Wavelets - 2D Discrete Wavelet transform

- In two dimensions, a two-dimensional scaling function $\varphi(x, y)$ and three 2-D wavelets $\psi^H(x, y)$, $\psi^V(x, y)$, $\psi^D(x, y)$ are required.
- Scaling function $\varphi(x, y) = \varphi(x)\varphi(y)$
- Seprable, “directionally sensitive” wavelets
$$\psi^H(x, y) = \psi(x)\varphi(y)$$
$$\psi^V(x, y) = \varphi(x)\psi(y)$$
$$\psi^D(x, y) = \psi(x)\psi(y)$$
- These wavelets measure functional variations intensity changes in images along different directions: columns, rows, diagonals

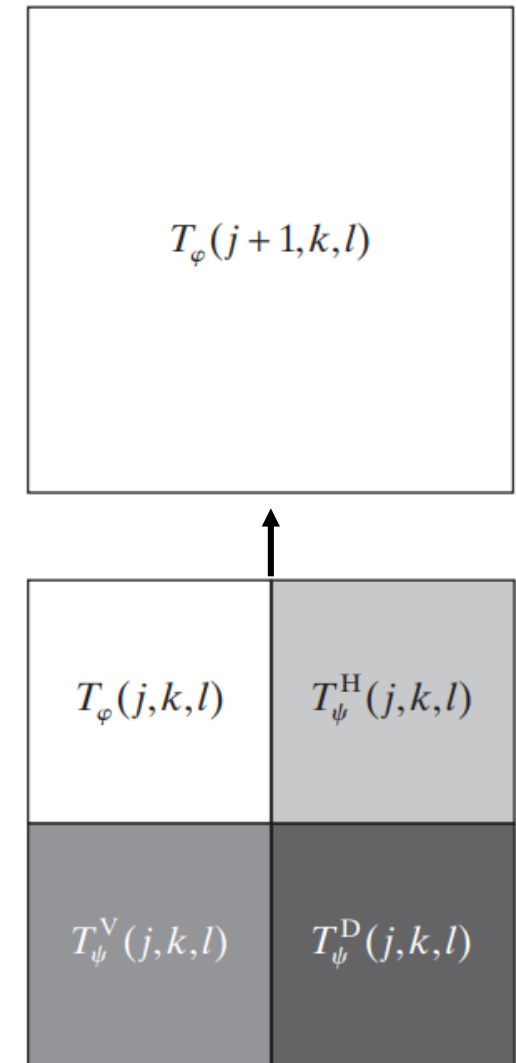
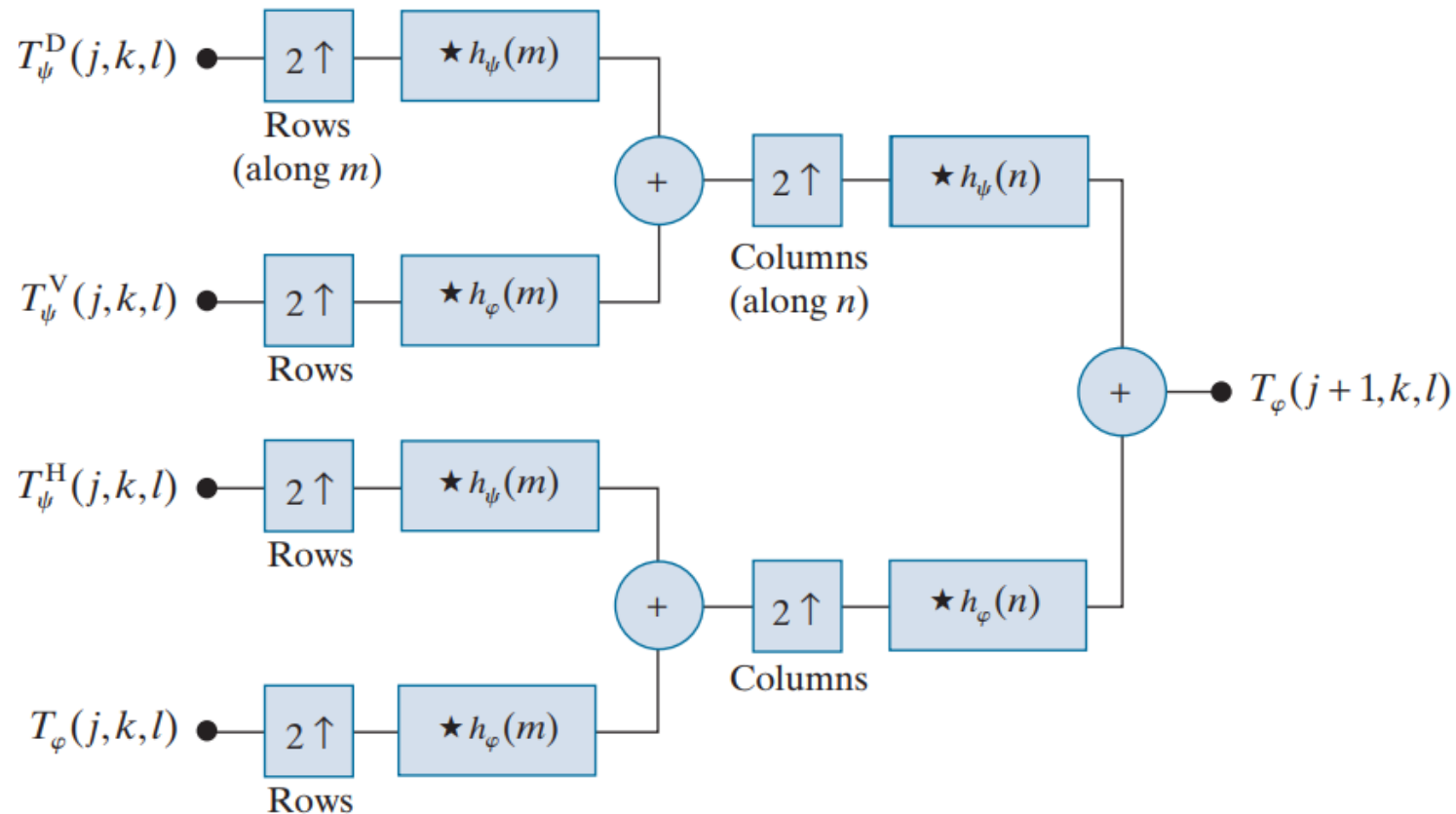
4.6 Multiscale Image Decompositions and Wavelets - 2D Discrete Wavelet transform

- Fast wavelet transform
- m and n are variables of convolution



4.6 Multiscale Image Decompositions and Wavelets - 2D Discrete Wavelet transform

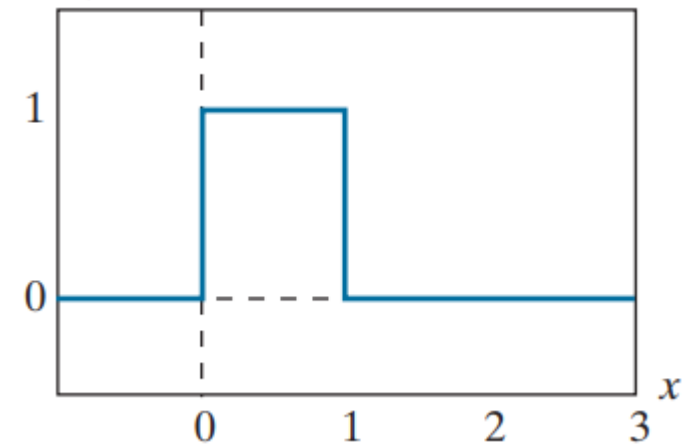
- Inverse fast wavelet transform
- m and n are dummy variables of convolution



4.6 Multiscale Image Decompositions and Wavelets - 2D Discrete Wavelet transform

- Example
- the unit-height, unit-width scaling function $\varphi(x) = \begin{cases} 1 & \text{if } 0 \leq x < 1 \\ 0 & \text{otherwise} \end{cases}$

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[33 33 32 33 33 33 31 29]
[34 33 32 33 34 33 29 28]
[32 31 30 31 33 31 28 27]
[30 29 28 29 31 30 28 27]
[29 28 28 28 30 30 28 25]
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Summary

- Image transforms
- Discrete Hartley transform
- Discrete Cosine transform
- Walsh-Hadamard Transforms
- Wavelet transform
- Multiscale Image Decompositions and Wavelets
- Machine learning based Image and Video analysis